Test Plan & Results

1. Test Plan

Test	Method	Expected Result
Ultrasonic sensor range	Move hand from 2cm to 30cm above sensor	Motor 1 responds and changes direction according to hand height
Motor 1 direction response	Place hand at low, mid and high positions	Motor moves up when hand is high, down when hand is low
Motor 2 continuous rotation	Power on system	Motor 2 spins continuously without stopping
Push button	Press button once	Turns on Laser
LDR accuracy	Switch on laser to see if values are detected	LDR value shows “1” every time beam is detected
Full system together	Run all components simultaneously	All components work without interfering with each other

2. User Testing

Testing was conducted with classmates who had no prior knowledge of the project setup or code. This ensured unbiased interaction with the model.

Tasks given to users:

- Press the push button
- Move your hand above the sensor and try to control the disk height
- Try to make the disk go as high as possible then as low as possible

3. Results

Test	Outcome	Pass/Fail

Ultrasonic sensor range	**Sensor Stopped working(hardware issue)**	**Fail**
Motor 1	**Working up and down**	**Pass**
Motor 2	**Rotation continuous Working**	**Pass**
Push Button	**Turns on laser**	**Pass**
LDR Accuracy	**Reads only when laser hits**	**Pass**
Full System together	**Only ultrasonic wasn't working. The DC motor 1 wasn't able to match the ultrasonic values**	**Pass (Only few components failed)**

4. Observations

- The system needed better coding to work as a whole as a whole.
- Some components failed last minute which wasn't expected.
- Users were intrigued by the aesthetics and mechanism of the Linear actuator.

6. Improvements Suggested

- Better code for more accuracy between ultrasonic readings and dc motor rotation